```
# Train.py
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
import joblib

df=pd.read_csv("iris.csv")

x=df.drop(columns=["Id","Species"])
y=df["Species"]

x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)

models={
"KNN":KNeighborsClassifier(),
"SVM":SVC(),
"Decision Tree":DecisionTreeClassifier(),
"Random Forest":RandomForestClassifier()
}

best_model=None
best_accuracy=0
best_name=""

print("Model Accuracy:\n")

for name,model in models.items():
```

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    model.fit(x_train,y_train)
    y_pred=model.predict(x_test)
    accuracy=accuracy_score(y_test,y_pred)
    print(f"{name}:{accuracy*100:.2f}%")
    if accuracy>best_accuracy:
        best_accuracy=accuracy
        best_model=model
        best_name=name

print(f"\nBest Model:{best_name}({best_accuracy*100:.2f}%)")

joblib.dump(best_model,"best_model.pkl")
print("Model saved as best_model.pkl")

# Test.py
import tkinter as tk
import joblib
import pandas as pd

try:
    model=joblib.load("best_model.pkl")
except:
    print("Please run train.py first")

df=pd.read_csv("iris.csv")

def predict():
    try:
        values=[
        float(sl.get()),
        float(sw.get()),
```

```
        float(pl.get()),
        float(pw.get())
        ]

        prediction=model.predict([values])[0]
        result.config(text=f"Prediction:{prediction}",fg="green")

    except:
        result.config(text="Enter valid values",fg="red")

root=tk.Tk()
root.title("Iris Prediction")
root.geometry("400x450")
root.configure(bg="black")

tk.Label(root,text="Iris Flower Predictor",font=("Arial",16,"bold"),bg="black",fg="white").pack(pady=10)

tk.Label(root,text="Sepal Length",font=("Arial",11),bg="black",fg="white").pack()
sl=tk.Entry(root,font=("Arial",11))
sl.pack(pady=5)

tk.Label(root,text="Sepal Width",font=("Arial",11),bg="black",fg="white").pack()
sw=tk.Entry(root,font=("Arial",11))
sw.pack(pady=5)

tk.Label(root,text="Petal Length",font=("Arial",11),bg="black",fg="white").pack()
pl=tk.Entry(root,font=("Arial",11))
pl.pack(pady=5)

tk.Label(root,text="Petal Width",font=("Arial",11),bg="black",fg="white").pack()
pw=tk.Entry(root,font=("Arial",11))
```

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pw.pack(pady=5)


tk.Button(root,text="Predict",command=predict,bg="green",fg="white",font=("Arial",12,"bold"),padx=10,pady=5).pack(pady=15)


result=tk.Label(root,text="",font=("Arial",12,"bold"),bg="black",fg="yellow")

result.pack()

root.mainloop()
```
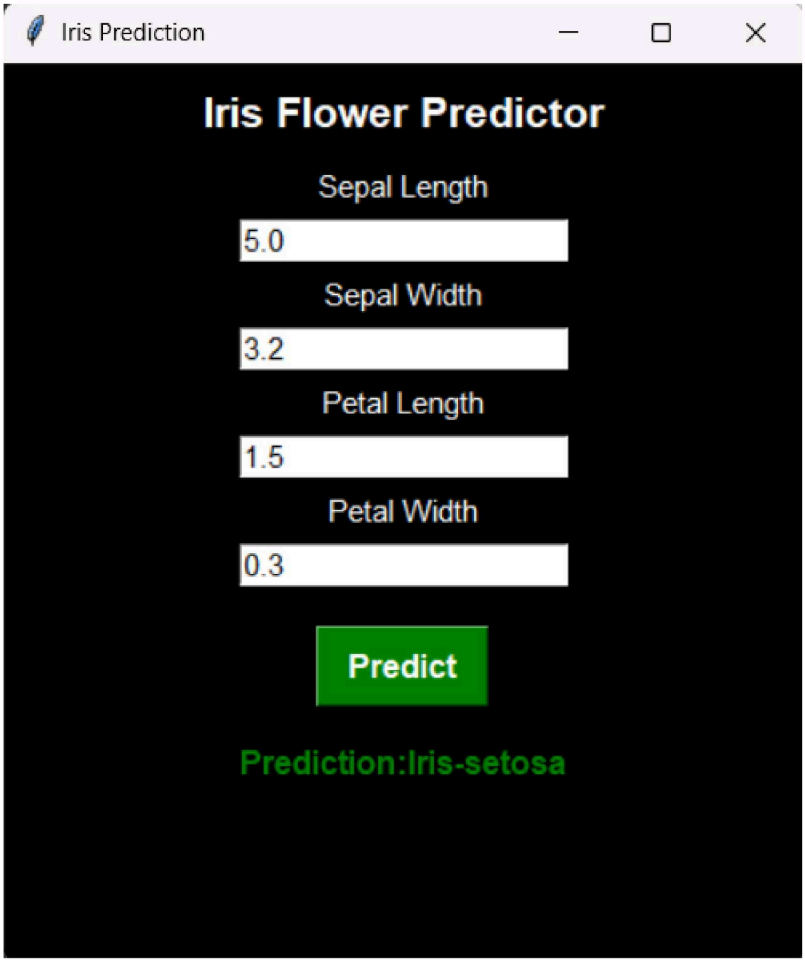

Output:

```
(.venv) PS D:\Internship\Day49> python task.py
Model Accuracy:

KNN:100.00%
SVM:100.00%
Decision Tree:100.00%
Random Forest:100.00%

Best Model:KNN(100.00%)
Model saved as best_model.pkl
```